King Fahd University of Petroleum and Minerals
Department of Mathematics
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Math 101 Final Exam Term 241

Original exam date: not printed. Practice version: questions and choices shuffled.
Net time allowed: 120 minutes

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1. The function $f(x) = \dfrac{\sqrt{x-2}}{x^2-3x-4}$ is continuous on

(a) $(-\infty,-1)\cup(-1,4)\cup(4,\infty)$

(b) $(-\infty,-1)\cup(-1,2]$

(c) $[2,4)\cup(4,\infty)$

(d) $(-1,4)$

(e) $[2,\infty)$

2. The graph of the function $f(x) = 8x^{1/3} - x^{4/3}$ has

(a) two local minima only

(b) neither local minimum nor local maximum

(c) one local minimum and one local maximum

(d) one local maximum only

(e) one local minimum only

3. If $y = Ax + B$ is an equation of the tangent line to the curve $y = e^{\sinh(2x)}$ at the point $(0,1)$, then $AB =$

(a) 2

(b) 1

(c) -3

(d) 3

(e) -4

4. The graph of the function $g(x) = x^4 e^{-2x}$ is

(a) decreasing on $(0,2)$

(b) increasing on $(0,\infty)$

(c) increasing on $(0,2)$

(d) decreasing on $(-\infty,2)$

(e) increasing on $(2,\infty)$

5. The graph of the function $f(x) = x^4 - 4x^3 + 2$ has

(a) no inflection points

(b) two inflection points

(c) four inflection points

(d) three inflection points

(e) one inflection point

6. $\lim_{x\to 0^+} \dfrac{2 - 3\ln x}{e^{2/x}} =$

(a) ∞

(b) 1

(c) 2

(d) $\dfrac{3}{2}$

(e) 0

7. What is the minimum area (cm^2) of the triangle formed in the first quadrant by the x-axis, the y-axis and a line through the point $(1, 2)$?

(a) 2

(b) 4

(c) 6

(d) 8

(e) 10

8. $\lim_{x\to\infty} \left[\cos\left(\dfrac{3}{x}\right)\right]^{2x^2} =$

(a) e^{-9}

(b) 1

(c) e^{-3}

(d) ∞

(e) e^{-6}

9. If $f'(x) = e^x + \dfrac{1}{\sqrt{x}}$ and $f(1) = e$, then $f(4) =$

(a) $e^4 - 3$

(b) $e^4 + 6$

(c) $e^4 - 4$

(d) $e^4 - 2$

(e) $e^4 + 2$

10. If $\tanh x = \dfrac{-2}{3}$, then $\sinh x =$

(a) $-\dfrac{2}{\sqrt{5}}$

(b) $-\dfrac{3}{\sqrt{5}}$

(c) $\dfrac{2}{\sqrt{5}}$

(d) $\dfrac{5}{\sqrt{5}}$

(e) $\dfrac{1}{\sqrt{5}}$

11. If (a, b) and (c, d) are the points at which the graph of $f(x) = \ln\left|\dfrac{x^2+5}{x+2}\right|$ has a horizontal tangent line, then $2ac =$

(a) 15

(b) 18

(c) -6

(d) -10

(e) 12

12. If $y = e^x \sin x$, then $y'' - 2y' + 2y =$

(a) $e^x \cos x$

(b) 0

(c) $3e^x \sin x$

(d) $e^x(\cos x - 2\sin x)$

(e) $-2e^x \sin x$

13. Two lines are tangent to the curve $y = 4x - x^2$ and both pass through the point $\left(\frac{5}{2}, 6\right)$. The product of the slopes of these two lines is equal to

(a) $-\frac{5}{8}$

(b) $-\frac{3}{4}$

(c) -10

(d) -12

(e) -8

14. If $f(x) = (2x-1)^3(3-2x)^5$, then $f'(x) =$

(a) $4(7-8x)(2x-1)^2(3-2x)^4$

(b) $2(9-10x)(2x-1)^2(3-2x)^4$

(c) $2(4-2x)(2x-1)^2(3-2x)^4$

(d) $(14-15x)(2x-1)^2(3-2x)^4$

(e) $4(7-2x)(2x-1)^2(3-2x)^4$

15. If $f(t) = 2t^3 + 3\cos t$, then $f'(t) =$

(a) $2t^2 - 3\cos t$

(b) $3t^2 - 3\sin t$

(c) $2t^3 + 3\cos t$

(d) $6t^2 - 3\sin t$

(e) $6t^2 + 3\sin t$

16. The graph of the function $f(x) = x^2 - \frac{1}{x}$ is

(a) concave up on $(0, 1)$

(b) concave down on $(-\infty, 0)$

(c) concave up on $(1, \infty)$

(d) concave down on $(1, \infty)$

(e) concave up on $(0, \infty)$

17. Which one of the following statements is TRUE? (V.A: vertical asymptote(s); H.A: horizontal asymptote(s))

(a) The graph of $f(x) = \dfrac{\sin(2x)}{x}$ has no V.A.

(b) Every rational function has at least one V.A.

(c) Every rational function has a slant asymptote

(d) Every rational function has two H.A.

(e) The graph of $f(x) = \dfrac{\sin(2x)}{x}$ has no H.A.

18. The absolute minimum value of $f(x) = x \ln x$ on $(0, \infty)$ is equal to

(a) $-\dfrac{2}{e}$

(b) $-\dfrac{1}{e}$

(c) $\dfrac{3}{e^2}$

(d) $-\dfrac{2}{e^2}$

(e) $\dfrac{1}{e^2}$

19. Question 7 of the original exam is omitted (missing from the published exam file (page 4 was not published)).

20. Question 8 of the original exam is omitted (missing from the published exam file (page 4 was not published)).

Answer Key

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Q	Ans	Brief explanation
1	**(c)**	Need $x \geq 2$ and $x \neq 4, -1$: $[2,4) \cup (4,\infty)$.
2	**(d)**	$f' = \frac{4(2-x)}{3x^{2/3}}$: no sign change at 0, $+ \to -$ at 2: one local max only.
3	**(a)**	$y'(0) = 2$, line $y = 2x + 1$, so $AB = 2 \cdot 1 = 2$.
4	**(c)**	$g' = 2x^3 e^{-2x}(2 - x) > 0$ exactly on $(0, 2)$: increasing there.
5	**(b)**	$f'' = 12x(x - 2)$ changes sign at $x = 0$ and $x = 2$: two inflection points.
6	**(e)**	L'Hopital: $\frac{3}{2} \cdot \frac{x}{e^{2/x}} \to 0$.
7	**(b)**	$A = \frac{x^2}{x-1}$, $A' = 0$ at $x = 2$: minimum area 4.
8	**(a)**	$\ln y \to -9$ by two applications of L'Hopital, so the limit is e^{-9}.
9	**(e)**	$f = e^x + 2\sqrt{x} + C$, $C = -2$, so $f(4) = e^4 + 2$.
10	**(a)**	$\operatorname{sech}^2 x = \frac{5}{9}$, $\cosh x = \frac{3}{\sqrt{5}}$, $\sinh x = -\frac{2}{\sqrt{5}}$.
11	**(d)**	$f' = \frac{(x+5)(x-1)}{(x^2+5)(x+2)} = 0$ at $x = -5, 1$, so $2ac = -10$.
12	**(b)**	$y' = e^x(\sin x + \cos x)$, $y'' = 2e^x \cos x$, so the combination is 0.
13	**(e)**	Tangency at $t = 1, 4$ gives slopes 2 and -4, product -8.
14	**(a)**	Product and chain rules: bracket $6(3-2x) - 10(2x-1) = 28 - 32x = 4(7-8x)$.
15	**(d)**	Power rule and $(\cos t)' = -\sin t$: $f' = 6t^2 - 3\sin t$.
16	**(c)**	$f'' = \frac{2(x^3-1)}{x^3} > 0$ for $x > 1$: concave up on $(1,\infty)$.
17	**(a)**	$\lim_{x\to 0} \frac{\sin 2x}{x} = 2$ (no V.A.); $y = 0$ is an H.A.; $\frac{1}{x^2+1}$ refutes the rest.
18	**(b)**	$f' = 1 + \ln x = 0$ at $x = e^{-1}$ (min): $f = -\frac{1}{e}$.
19	omitted	Omitted.
20	omitted	Omitted.